

CS 630: Software Testing and Quality Assurance

6-1 Project One: Software Testing and Quality Assurance Report

Malgorzata Debska

Southern New Hampshire University

Dr. Ben Torrales

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CS 630 Project One: Detailed Issues and Recommendations – Excelon Development

Executive Quality Evaluation

Based on the executed evidence, Excelon Development is conditionally ready for the PR push.

Core service information is reachable; Testing, Placement, Writing, Consulting, and Software offerings are discoverable, and recent blog content is available. No confirmed Critical or High functional failure was found in executed core paths. The recommendation is “Proceed after targeted pre-launch verification” because responsive/cross-browser behavior, accessibility details, contact delivery, and measured performance were not exhaustively demonstrated.

Final-stage QA must evaluate coverage and residual risk, not only observed bugs. An effective QA report should communicate meaningful status, findings, and risk in a form stakeholders can use for decisions (Beregnyaya, 2020). This report therefore separates confirmed observations from unverified risks and assigns actions according to likely PR impact.

Prioritized Improvements

Priority	Finding / Risk	Business Impact	Recommendation / Owner	Release Consequence
P1 – High	Cross-browser and real-device visual coverage is incomplete.	A layout/menu failure seen by PR-driven visitors could directly harm credibility or block lead generation.	QA: execute critical smoke suite on current Chrome, Safari, Firefox, Edge and at least one iOS/Android phone before campaign.	Block PR launch if a Critical/High journey failure is found; otherwise document results.
P1 – High	Accessibility-oriented verification is incomplete; some client images appear generically in retrievable structure.	Missing/weak alternatives or keyboard problems can exclude users and reduce professional quality.	Developer + QA: inspect accessibility tree, meaningful image alternatives, keyboard order, visible focus, headings, and link purpose; remediate confirmed failures.	High-severity navigation/accessibility barrier requires fix or formal acceptance.
P1 – High	No objective load/performance baseline was established during this limited run.	PR traffic can increase load; slow pages reduce trust and conversion even when functions technically work.	QA/DevOps: capture page-load/Core Web Vitals baseline and test critical pages under representative traffic; establish alerting.	Severe degradation or instability blocks campaign until mitigated.
P2 – Medium	Homepage repeats Testing, Placement, and	Repetition may make the page feel less polished	Product/content owner: review intended layout	Does not block release unless it causes

	Writing service summaries.	and can dilute the primary message for first-time visitors.	and remove accidental duplication if not purposeful.	layout/accessibility problems.
P2 – Medium	Contact/conversion paths should receive a final end-to-end production check without creating unwanted submissions.	A broken lead path would undermine the main business purpose of PR traffic.	QA + business owner: execute controlled contact-form validation, required-field/error checks, confirmation behavior, and receipt verification.	Blocking if valid prospects cannot submit or messages are not delivered.
P3 – Low	Content/copy should receive a final consistency pass across older and newer pages.	Typos, inconsistent capitalization, outdated claims, or inconsistent CTA wording can reduce credibility.	Content owner: proofread service, About, team, footer, and recent blog landing pages; verify dates and external links.	Track for cleanup; normally non-blocking.

Evaluation Against QA Standards

ISO/IEC/IEEE 29119-1 supports consistent terminology, so leadership can distinguish a confirmed defect, an anomaly requiring investigation, and an untested risk. This report preserves those distinctions.

ISO/IEC/IEEE 29119-3 is the documentation reference and provides templates usable across projects and lifecycle models (ISO, 2021a). This project applies that structure through scope, cases/charters, expected and actual results, logs, findings, completion assessment, and release recommendation. Structured documentation also improves traceability and communication (Sokurenko, 2025).

ISO/IEC/IEEE 29119-4 is applied as test-design guidance, consistent with instructor feedback; it defines techniques for test design and implementation (ISO, 2021b). Cases were therefore derived from business-critical user paths and representative conditions rather than random page checks.

Context-driven testing is appropriate because Excelon itself describes its approach as risk-oriented and context-driven. Limited effort is therefore concentrated on public visitor experience and reputational risk rather than irrelevant commerce/store-locator scenarios from the sample plan.

Readiness for the Public-Relations Push

Release status: CONDITIONAL GO. Evidence is strongest for content reachability and service navigation and weakest for device/browser visuals, accessibility behavior, contact delivery, and performance under increased traffic. Leadership should require a short verification gate before final approval. The PR launch should stop for any Critical defect; a High defect blocking service information/contact; a severe supported-mobile layout failure; a critical-journey accessibility barrier; or instability likely to affect campaign traffic. Medium/Low issues may be accepted only with documented impact, owner, remediation date, and residual-risk approval.

Findings and Open Issues

Confirmed execution showed the homepage and major service/content paths were available: Testing includes risk-management content and contact; Consulting explains scope/delivery; Writing exposes recent publications; Placement describes staffing; and Blog includes recent dated content. These support service discovery and credibility.

Open items require environment-specific evidence: verify meaningful image alternatives, confirm homepage repetition is intentional, and complete cross-browser/mobile, keyboard/accessibility, controlled contact-submission, and performance checks.

Risks and Items Untested

Residual risks include real iOS/Android rendering, older supported environments, screen-reader and keyboard behavior, zoom/high-contrast behavior, controlled contact delivery, PR-load behavior, third-party outages, deeper security testing, and backend/infrastructure behavior. These are explicitly untested or partially tested - not pass.

Suggested Next Steps and Governance

Next steps are run a risk-based smoke suite across supported browsers/devices; complete accessibility-oriented checks; authorize and verify one controlled contact submission; capture a performance baseline/monitoring; then rerun regression smoke and issue a final release report. QA owns evidence, severity, and the release recommendation; developers own technical fixes; content/product owners decide intentional presentation changes; and business leadership owns final PR approval and documented risk acceptance.

Conclusion

Excelon demonstrates acceptable core functional quality in the executed scope with no confirmed release-blocking defect in primary information/service paths. A conditional decision is still appropriate: targeted browser/device, accessibility, contact, and performance verification should precede the PR campaign. This gives leadership a defensible record of what passed, what remains uncertain, what blocks launch, and who owns each action.

References

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